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Total No. of Questions : 13

Total No. of Pages : 03

B.Pharma (Sem-8)

BIOSTATISTICS AND RESEARCH METHODOLOGY

Subject Code : BP-801T

M.Code : 79764

Date of Examination : 01-06-2023

Time : 3 Hrs.

Max. Marks : 75

INSTRUCTIONS TO CANDIDATES :

1. SECTION-A is COMPULSORY consisting of TEN questions carrying TWO marks each.
2. SECTION-B contains THREE questions carrying TEN marks each and student has to attempt any TWO questions.
3. SECTION-C contains NINE questions carrying FIVE marks each and student has to attempt any SEVEN questions.

SECTION-A

1. **Answer Briefly :**

a) Define Coefficient of correlation and coefficient of determination.

b) Give applications of Biostatistics.

c) Distinguish between paired and independent Student's t test.

d) Define Null Hypothesis and Level of Significance.

e) What do you understand by cohort studies?

f) Give major applications of R-Online statistical software in clinical trials.

g) Differentiate between parametric and non-parametric tests.

h) Write down advantages of Factorial Designs.

i) What are statistical errors?

j) Define power of a study.

IV/5.6.23

SECTION-B

2. Discuss the need for design of experiments. Explain various experimental design techniques.

3. a) In elevated plus maze model, mean number of entries made by mice in open arms after treatment with vehicle (control) and test drug were recorded. Analyze whether significant difference exists between control and test drug.

Control	Test Drug
5.2	14.2
4.8	13.8
5.3	12.8
3.6	11.6
4.2	13.2
4.0	12.8
4.2	14.4
3.8	15.2
2.8	11.8
	12.4

b) Write a note on 'ANOVA'.

4. a) Given the bivariate distribution shown in Table, in which both X and Y are random variables

i) Compute the correlation coefficient;

ii) Test the sample correlation coefficient for statistical significance;

Number	X	Y
1	12	10
2	14	18
3	20	22
4	33	18
5	23	25
6	19	20
7	22	19

b) Explain various 'Measures of Dispersion'.

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5. Write a note on Central Composite Design.

6. What are factorial Designs? How 2^3 and 2^4 designs help in analysis of experiments?

7. Explain blocking systems for two level factorials.

8. Calculate mean, median and mode from the following data :

Marks	No. of Students
10-25	6
25-40	33
40-55	44
55-70	12
70-85	3
85-100	2

9. Write a note on Kruskal Wallis test.

10. Explain various steps in Report Writing.

11. What are different types of sampling? Explain non-probability sampling methods.

12. What is Multiple Regression? Describe its significance in analysis of clinical variables.

13. Write a detailed note on SPSS statistical software.

...sure of identity by writing Mobile No. or Marking of passing request on any paper of Answer Sheet will lead to UMC against the Student.